

Mohammad Tawhidul Hasan Bhuiyan

✉ mb5332@columbia.edu | [in](#) Tawhid Bhuiyan | [mthbhuiyan.github.io](#)

RESEARCH INTEREST

Compilers, Hardware/Software Co-Design

EDUCATION

Columbia University

PhD in Electrical Engineering

New York, USA

Sept. 2024 – May 2030 (expected)

Bangladesh University of Engineering and Technology (BUET)

Masters of Science in Computer Science and Engineering

Palashi, Dhaka, Bangladesh

July 2021 – Aug. 2024

Bangladesh University of Engineering and Technology (BUET)

Bachelor of Science in Computer Science and Engineering

Palashi, Dhaka, Bangladesh

Mar. 2016 – Feb. 2021

🎓 *Class rank: 1st*

EMPLOYMENT

Columbia University

Graduate Research and Teaching Assistant

New York, USA

Sept. 2024 – Present

Bangladesh University of Engineering and Technology (BUET)

Lecturer, Department of CSE

Dhaka, Bangladesh

Sept. 2021 – Aug. 2024

PUBLICATIONS

- **Tawhid Bhuiyan**, Sumya Hoque, Angelica Moreira, Tanvir Ahmed Khan, **Wax: Optimizing Data Center Applications With Stale Profile**, *ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, Volume 2, Association for Computing Machinery (ACM), 2026
- Ryan Piersma, **Tawhid Bhuiyan**, Tanvir Ahmed Khan, and Simha Sethumadhavan, **Reverse Engineering DVFS Mechanisms**, *2025 IEEE International Symposium on Hardware Oriented Security and Trust (HOST)*, San Jose, CA, USA, 2025
- **Mohammad Tawhidul Hasan Bhuiyan**, Muhammad Rashed Alam and M. Sohel Rahman, **Computing the Largest Common Almost-Increasing Subsequence**, *Theoretical Computer Science*, Volume 930, 2022, Pages 157-178, ISSN 0304-3975
- **Mohammad Tawhidul Hasan Bhuiyan**, Irtesam Mahmud Khan, Sheikh Saifur Rahman Jony, Renee Robinson, Uyen-Sa D. T. Nguyen, David Keellings, M. Sohel Rahman, and Ubydul Haque, **The Disproportionate Impact of COVID-19 among Undocumented Immigrants and Racial Minorities in the US**, *International Journal of Environmental Research and Public Health*, Volume 18(23): 12708, 2021, PubMedID 34886437, ISSN 1660-4601

INVITED TALKS & PRESENTATIONS

- June 27, 2026: Presented *Enabling Energy Efficiency for Distributed Machine Learning* at the ASTRA-sim Tutorial (ISCA 2026)
- Mar. 9, 2026: Presented *WAX: Optimizing Data Center Applications With Stale Profile* to Google
- Feb. 5, 2026: Presented *WAX: Optimizing Data Center Applications With Stale Profile* to Meta

TECHNICAL SKILLS

Architectural Simulators: Accel-sim, Astra-sim

Compiler & Systems: LLVM, BOLT

Performance Tools: perf, NVIDIA Nsight, NVML, CUPTI, PyTorch Profiler

Languages: C/C++, Python, Java, SQL, Bash, JavaScript

Libraries: Polars, Numba, PyTorch, Scikit-Learn, Tensorflow, Pandas, NumPy, Matplotlib

AWARDS

- BUET Alumni Association Award for securing top position in the department
- Dean's Honor List, BUET in all 8 undergraduate semesters
- University Merit Scholarship, BUET

TEACHING & MENTORSHIP

Columbia University

Mentor

- Mentored Amir Zarandi (Undergraduate student, Columbia University) in conducting original research on GPU energy modeling, resulting in a co-authored paper that is under submission.
- Mentored 15 students across 4 technical projects. Guided teams through the design, implementation, and debugging of complex systems. Some key projects include:
 - *Fine-grained GPU DVFS*: Led the students to update Accel-sim to allow simulating and evaluating new DVFS techniques for GPU cores.
 - *HBM DVFS & Co-Optimization*: Led the investigation into HBM idle power, utilizing DRAMsim3 and Ramulator2 to propose co-optimized DVFS mechanisms for GPU compute and memory.
 - *Power-Aware Collective Communication*: Guided the integration of different ML models into Astra-sim to easily simulate different model architectures.

Bangladesh University of Engineering and Technology (BUET)

Lecturer

- Developed curriculum and delivered lectures for core undergraduate courses:
 - *Operating Systems*: Led hands-on sessions on modifying open-source kernels; taught memory virtualization techniques and file systems.
 - *Database Systems*: Instructed on schema design, normalization theory, and the internal architecture of DB storage systems.

COMMUNITY SERVICES

Organizing Committee Member of NSysS 2021, 2022, 2023

REFERENCES

Tanvir Ahmed Khan

Assistant Professor

Department of Electrical Engineering

Columbia University

✉ tk3070@columbia.edu